CSE402: Optimisation

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1 Lecture 01

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Mathematical language of this course is linear algebra (of real vector spaces) and multivariable calculus. In this lecture, we will give a brief review of linear algebra.

§1.1 Review on linear algebra

Notation. Let $a_1, a_2, \ldots, a_n$ be a finite sequence of real numbers. We will denote the sum

$$S = a_1 + a_2 + \cdots + a_n \tag{1.1}$$

by

$$S = \sum_{k=1}^{n} a_k. \tag{1.2}$$

We have the following useful identities:

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}, \tag{1.3}$$

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}. \tag{1.4}$$

§1.1.1 Vectors

Definition 1.1.1 ($\mathbb{R}$-vector space)**.** Let V be a non-empty set. The set V equipped with two binary operations

$$+ : V \times V \to V$$

called *vector addition*, and

$$\cdot : \mathbb{R} \times V \to V$$

called *scalar multiplication* is called a **vector space over** $\mathbb{R}$ or **$\mathbb{R}$-vector space** if for all $u, v, w \in V$ and all $a, b \in \mathbb{F}$, the following axioms hold:

1. **Closure under addition:**
$$u + v \in V \tag{1.5}$$

2. **Commutativity of addition:**
$$u + v = v + u \tag{1.6}$$

3. **Associativity of addition:**
$$(u + v) + w = u + (v + w) \tag{1.7}$$

4. **Existence of zero vector:** There exists a vector $0 \in V$ such that

$$u + 0 = u \tag{1.8}$$

5. **Existence of additive inverse:** For every $u \in V$, there exists $-u \in V$ such that

$$u + (-u) = 0 \tag{1.9}$$

6. **Closure under scalar multiplication:**

$$au \in V \tag{1.10}$$

7. **Distributivity of scalar multiplication over vector addition:**

$$a(u + v) = au + av \tag{1.11}$$

8. **Distributivity of scalar multiplication over field addition:**

$$(a + b)u = au + bu \tag{1.12}$$

9. **Associativity of scalar multiplication:**

$$a(bu) = (ab)u \tag{1.13}$$

10. **Identity element of scalar multiplication:**

$$1u = u \tag{1.14}$$

Hence, $(V, +, \cdot)$ is called a vector space over the field $\mathbb{R}$. Elements of V are called **vectors** and elements of $\mathbb{R}$ are called **scalars**.

Convention. From now on, vector space means $\mathbb{R}$-vector space, unless stated otherwise.

In this course, we will work with only the following special example of $\mathbb{R}$-vector space.

Example 1.1.2

Let

$$V \coloneqq \mathbb{R}^n = \{[x_1, x_2 \ldots, x_n]^T \mid x_1, x_2, \ldots, x_n \in \mathbb{R}\}. \tag{1.15}$$

For $[x_1, x_2 \ldots, x_n]^T, [y_1, y_2 \ldots, y_n]^T \in V$ and $\alpha \in \mathbb{R}$ define addition $(+)$ and scalar multiplication $(\cdot)$, respectively as follows:

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{bmatrix}, \tag{1.16}$$

$$\alpha \cdot \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} \alpha x_1 \\ \alpha x_2 \\ \vdots \\ \alpha x_n \end{bmatrix}. \tag{1.17}$$

Then it is an easy check that $V = \mathbb{R}^n$ forms an $\mathbb{R}$-vector space.

§1.1.2 Linear transformations

In this course, we are mostly concerned with following special functions between vector spaces:

Not sure about that

Definition 1.1.3 (Linear transformation)**.** Let V, W be two vector space. A **linear transformation** from V to W is a function $T : V \to W$ such that for all $\mathbf{x}, \mathbf{y} \in V$ and $\alpha \in \mathbb{R}$ the following holds true:

- **Additivity.**
$$T(\mathbf{x} + \mathbf{y}) = T(\mathbf{x}) + T(\mathbf{y}). \tag{1.18}$$
- **Homogeneity.**
$$T(\alpha \cdot \mathbf{x}) = \alpha \cdot T(\mathbf{x}). \tag{1.19}$$

Need to add visualisation of additivity and homogeneity property of a linear transformation.

Example 1.1.4

Let $V \coloneqq \mathbb{R}^n, W \coloneqq \mathbb{R}^m$. An $m \times n$ matrix M then defines a linear transformation $M : \mathbb{R}^n \to \mathbb{R}^m$[a] as follows:

For $\mathbf{x} \in \mathbb{R}^n$,

$$M(\mathbf{x}) \coloneqq M\mathbf{x}. \tag{1.20}$$

Here, M is an $m \times n$ matrix, $\mathbf{x}$ can is an $n \times 1$ matrix. So $M\mathbf{x}$ is an $m \times 1$ matrix. Therefore, M is indeed a function from $\mathbb{R}^n$ to $\mathbb{R}^m$.

Furthermore, for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$, the following holds by the properties of matrix multiplication:

$$M(\mathbf{x} + \mathbf{y}) = M\mathbf{x} + M\mathbf{y}, \tag{1.21}$$

$$M(\alpha\mathbf{x}) = \alpha M\mathbf{x} \tag{1.22}$$

Hence, M is indeed a linear transformation.

[a] Here, we are denoting the linear transformation determined by the matrix M by the same letter M using abuse of notation.

Surprising fact is that an arbitrary linear transformation also comes from a matrix. Before that, we need the idea of basis.

§1.1.3 Basis and dimension of $\mathbb{R}^n$

Consider the vector space

$$\mathbb{R}^2 = \left\{ [x, y]^T : x, y \in \mathbb{R} \right\}. \tag{1.23}$$

A natural question is: how many vectors are required to generate every vector in $\mathbb{R}^2$? Answer to this question is particularly useful in the context of linear transformations.

Suppose

$$L : \mathbb{R}^2 \to V$$

is a linear transformation. Because $\mathbb{R}^2$ contains infinitely many vectors, describing which vector maps to what under L would require an infinite amount of information.

However, suppose every vector of $\mathbb{R}^2$ can be generated by a finite collection of vectors

$$v_1, \ldots, v_n \tag{1.24}$$

by taking "linear combinations". Then, instead of knowing the value of L on every vector, we only need to know the values

$$L(v_1), \ldots, L(v_n). \tag{1.25}$$

Indeed, if

$$v = a_1 v_1 + \cdots + a_n v_n, \tag{1.26}$$

then by the linearity of L,

$$L(v) = L(a_1 v_1 + \cdots + a_n v_n) = a_1 L(v_1) + \cdots + a_n L(v_n). \tag{1.27}$$

Therefore, knowing the images of $v_1, \ldots, v_n$ completely determines the linear transformation L on the whole space.

This motivates the following question: What is the smallest collection of vectors required to generate all vectors of a vector space?

The answer leads us to the concepts of spanning sets, linear independence, basis, and dimension:

Definition 1.1.5 (Linear Independence)**.** A collection of vectors

$$\{v_1, v_2, \ldots, v_k\} \tag{1.28}$$

in a vector space V is called *linearly independent* if

$$a_1 v_1 + a_2 v_2 + \cdots + a_k v_k = 0 \tag{1.29}$$

implies

$$a_1 = a_2 = \cdots = a_k = 0. \tag{1.30}$$

If there exists a non-zero solution of the above equation, then the vectors are called *linearly dependent*.

Intuitively, if a list of vectors are linearly dependent, then at least one of them can be generated by rest of them.

Example 1.1.6

The vectors

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad \begin{bmatrix} 1 \\ -1 \end{bmatrix} \tag{1.31}$$

are linearly independent.

However,

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad \begin{bmatrix} 2 \\ 4 \end{bmatrix} \tag{1.32}$$

are linearly dependent because the second vector is a scalar multiple of the first.

Definition 1.1.7 (Span)**.** Let V be a vector space and let

$$S = \{v_1, v_2, \ldots, v_n\} \tag{1.33}$$

be a collection of vectors in V. The span of S, denoted by

$$\text{span}\{v_1, v_2, \ldots, v_n\}, \tag{1.34}$$

is the set of all possible linear combinations of the vectors in S. That is,

$$\text{span}\{v_1, v_2, \ldots, v_n\} = \{a_1 v_1 + a_2 v_2 + \cdots + a_n v_n : a_1, \ldots, a_n \in \mathbb{R}\}. \tag{1.35}$$

If

$$\text{span}\{v_1, v_2, \ldots, v_n\} = V, \tag{1.36}$$

then we say that $v_1, v_2, \ldots, v_n$ span the vector space V.

A basis is a minimal spanning set, that is, it contains enough vectors to generate the whole space, but no redundant vectors.

Example 1.1.8

Consider the vectors in $\mathbb{R}^2$

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}. \tag{1.37}$$

The span of these two vectors is

$$\text{span}\{v_1, v_2\} = \left\{ a \begin{bmatrix} 1 \\ 1 \end{bmatrix} + b \begin{bmatrix} 1 \\ -1 \end{bmatrix} : a, b \in \mathbb{R} \right\}. \tag{1.38}$$

Now,

$$a \begin{bmatrix} 1 \\ 1 \end{bmatrix} + b \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} a+b \\ a-b \end{bmatrix}. \tag{1.39}$$

Given any vector

$$\begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2, \tag{1.40}$$

choosing

$$a = \frac{x+y}{2}, \qquad b = \frac{x-y}{2} \tag{1.41}$$

gives

$$\begin{bmatrix} x \\ y \end{bmatrix} = a \begin{bmatrix} 1 \\ 1 \end{bmatrix} + b \begin{bmatrix} 1 \\ -1 \end{bmatrix}. \tag{1.42}$$

Therefore,

$$\text{span}\{v_1, v_2\} = \mathbb{R}^2. \tag{1.43}$$

Hence, the vectors v_1, v_2 span $\mathbb{R}^2$.

Definition 1.1.9 (Basis)**.** Let V be a vector space. A set of vectors

$$\mathcal{B} = \{v_1, v_2, \ldots, v_k\} \tag{1.44}$$

is called a *basis* of V if:

1. $v_1, v_2, \ldots, v_k$ span V, i.e.

$$V = \text{span}\{v_1, v_2, \ldots, v_k\}, \tag{1.45}$$

2. $v_1, v_2, \ldots, v_k$ are linearly independent.

Example 1.1.10

Consider the set of vectors

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}. \tag{1.46}$$

We claim that $\mathcal{B}$ is a basis of $\mathbb{R}^2$.
First, let

$$\begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2 \tag{1.47}$$

be an arbitrary vector. We want to express it as a linear combination of the vectors in $\mathcal{B}$. Suppose

$$\begin{bmatrix} x \\ y \end{bmatrix} = a \begin{bmatrix} 1 \\ 1 \end{bmatrix} + b \begin{bmatrix} 1 \\ -1 \end{bmatrix}. \tag{1.48}$$

Then,

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} a+b \\ a-b \end{bmatrix}. \tag{1.49}$$

Solving the equations

$$a + b = x, \qquad a - b = y, \tag{1.50}$$

we get

$$a = \frac{x+y}{2}, \qquad b = \frac{x-y}{2}. \tag{1.51}$$

Therefore, every vector in $\mathbb{R}^2$ can be written as a linear combination of the vectors in $\mathcal{B}$. Hence,

$$\text{span}(\mathcal{B}) = \mathbb{R}^2. \tag{1.52}$$

Also, the two vectors in $\mathcal{B}$ are linearly independent. Therefore, $\mathcal{B}$ is a basis of $\mathbb{R}^2$.
Moreover, the representation

$$\begin{bmatrix} x \\ y \end{bmatrix} = a \begin{bmatrix} 1 \\ 1 \end{bmatrix} + b \begin{bmatrix} 1 \\ -1 \end{bmatrix} \tag{1.53}$$

is unique. The scalars a and b are called the coordinates of the vector with respect to the basis $\mathcal{B}$.

§1.1.4 Dimension

The number of vectors contained in a basis gives the size of a vector space.

Theorem 1.1.11

Every basis of a finite-dimensional vector space contains the same number of vectors.

Proof. This proof was not shown in the class. Interested students may refer to [Axl24, p. 44]. □

is it examinable?

Definition 1.1.12 (Dimension)**.** The number of vectors in a basis of a vector space V is called the *dimension* of V, denoted by

$$\dim(V).$$

Example 1.1.13

Since every basis of $\mathbb{R}^2$ contains exactly two vectors,

$$\dim(\mathbb{R}^2) = 2.$$

§1.1.5 Basis and dimension of $\mathbb{R}^n$

The standard basis of $\mathbb{R}^n$ is

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \quad \ldots, \quad e_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}. \tag{1.54}$$

Every vector

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \tag{1.55}$$

can be written as

$$x_1 e_1 + x_2 e_2 + \cdots + x_n e_n. \tag{1.56}$$

Therefore,

$$\{e_1, e_2, \ldots, e_n\} \tag{1.57}$$

is a basis of $\mathbb{R}^n$.

Theorem 1.1.14

The dimension of $\mathbb{R}^n$ is

$$\dim(\mathbb{R}^n) = n. \tag{1.58}$$

§1.1.6 Eigenvectors and eigenvalues

Suppose we have a linear transformation $L : \mathbb{R}^n \to \mathbb{R}^m$. We are interested in finding a non-zero vector $\mathbf{x} \in \mathbb{R}^n$ such that

$$L\mathbf{x} = \lambda\mathbf{x}, \tag{1.59}$$

where $\lambda \in \mathbb{R}$.

Definition 1.1.15 (Eigenvector and eigenvalue)**.** $\mathbf{x}, \lambda$ (if exists) are called an **eigenvector** and **eigenvalue** for L. The $(\mathbf{x}, \lambda)$ is called an **eigenpair** for L.

Theorem 1.1.16

Let L be a linear transformation. Then $(\mathbf{x}, \lambda)$ is an eigen pair of L if and only if $\det(L - \lambda I) = 0$.

Proof. Suppose λ is an eigenvalue of L. Then there exists a non-zero vector $\mathbf{x} \in \mathbb{R}^n$ such that

$$L\mathbf{x} = \lambda\mathbf{x}. \tag{1.60}$$

Since $I\mathbf{x} = \mathbf{x}$, we have

$$L\mathbf{x} - \lambda I\mathbf{x} = 0, \tag{1.61}$$

or equivalently,

$$(L - \lambda I)\mathbf{x} = 0. \tag{1.62}$$

Since $\mathbf{x} \neq 0$, this homogeneous system has a non-trivial solution. Therefore, $L - \lambda I$ is singular. Hence,

$$\det(L - \lambda I) = 0. \tag{1.63}$$

Conversely, suppose $\det(L-\lambda I) = 0$. Then $L-\lambda I$ is singular, so there exists a non-zero vector $\mathbf{x} \in \mathbb{R}^n$ satisfying

$$(L - \lambda I)\mathbf{x} = 0. \tag{1.64}$$

Therefore,

$$L\mathbf{x} = \lambda\mathbf{x}. \tag{1.65}$$

Thus, λ is an eigenvalue of L, with corresponding eigenvector $\mathbf{x}$. □

Definition 1.1.17 (Eigenvalues of a matrix)**.** Let M be an $n \times n$ matrix. The set of all eigenvalues of M is denoted by

$$\text{eigs}(M) = \{\lambda_1, \lambda_2, \ldots, \lambda_n\}. \tag{1.66}$$

Here, the eigenvalues are listed with multiplicity.

Definition 1.1.18 (Definiteness of a matrix)**.** Let M be a real symmetric $n \times n$ matrix with eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$. Then M is called

1. **Positive definite (PD)** if all eigenvalues are strictly positive, i.e.,
$$\lambda_i > 0, \quad \forall i = 1, 2, \ldots, n. \tag{1.67}$$
2. **Negative definite (ND)** if all eigenvalues are strictly negative, i.e.,
$$\lambda_i < 0, \quad \forall i = 1, 2, \ldots, n. \tag{1.68}$$
3. **Positive semidefinite (PSD)** if all eigenvalues are non-negative, i.e.,
$$\lambda_i \geq 0, \quad \forall i = 1, 2, \ldots, n. \tag{1.69}$$
4. **Negative semidefinite (NSD)** if all eigenvalues are non-positive, i.e.,
$$\lambda_i \leq 0, \quad \forall i = 1, 2, \ldots, n. \tag{1.70}$$
5. **Indefinite (ID)** if M has both positive and negative eigenvalues.

Example 1.1.19

Let
$$L := \begin{bmatrix} -2 & 2 \\ 3 & -1 \end{bmatrix}. \tag{1.71}$$
We will find the eigenvalues and eigenvectors for L.

The eigenvalues are obtained by solving $\det(L - \lambda I) = 0$. Here,
$$\begin{aligned} \det(L - \lambda I) &= \begin{vmatrix} -2-\lambda & 2 \\ 3 & -1-\lambda \end{vmatrix} \\ &= (-2-\lambda)(-1-\lambda) - 6 \\ &= \lambda^2 + 3\lambda - 4 \\ &= (\lambda - 1)(\lambda + 4). \end{aligned} \tag{1.72}$$
Hence, the eigenvalues for L are $\lambda = 1$ and $\lambda = -4$.

For each eigenvalue, we find a non-zero vector $\mathbf{x} = [x_1, x_2]^T$ such that $(L - \lambda I)\mathbf{x} = 0$.

For $\lambda = 1$, we have
$$\begin{bmatrix} -3 & 2 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \tag{1.73}$$
Both equations give $3x_1 = 2x_2$, so $x_2 = \frac{3}{2}x_1$. Therefore, the eigenvectors corresponding to $\lambda = 1$ are precisely the vectors
$$\mathbf{x} = \begin{bmatrix} x_1 \\ \frac{3}{2}x_1 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ \frac{3}{2} \end{bmatrix}, \qquad x_1 \in \mathbb{R} \setminus \{0\}. \tag{1.74}$$
Choosing $x_1 = 2$, we obtain the eigenvector $\mathbf{x} = [2, 3]^T$.

For $\lambda = -4$, we have
$$\begin{bmatrix} 2 & 2 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \tag{1.75}$$

Both equations give $x_1 + x_2 = 0$, so $x_2 = -x_1$. Therefore, the eigenvectors corresponding to $\lambda = -4$ are precisely the vectors

$$\mathbf{x} = \begin{bmatrix} x_1 \\ -x_1 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \qquad x_1 \in \mathbb{R} \setminus \{0\}. \tag{1.76}$$

Choosing $x_1 = 1$, we obtain the eigenvector $\mathbf{x} = [1, -1]^T$.
Hence, $([2, 3]^T, 1)$ and $([1, -1]^T, -4)$ are eigenpairs for L.

Remark 1.1.20. Any nonzero scalar multiple of an eigenvector is also an eigenvector corresponding to the same eigenvalue.

Indeed, let $x \neq 0$ be an eigenvector of L corresponding to the eigenvalue λ. Then

$$Lx = \lambda x. \tag{1.77}$$

For any scalar $c \neq 0$, by linearity of L,

$$L(cx) = cLx = c\lambda x = \lambda(cx). \tag{1.78}$$

Hence, cx is also an eigenvector of L with the same eigenvalue λ.

2 Lecture 02

Date: 11/10/2026

§2.1 Review on linear algebra—continued

Exercise 2.1.1. Find eigenvalues of the matrix

$$\begin{bmatrix} 6 & 0 \\ 0 & 7 \end{bmatrix}, \tag{2.1}$$

and classify it in terms of eigenvalues.

Answer 2.1.2. Let M denote the matrix in (2.1). The eigenvalues are obtained by solving $\det(M - \lambda I) = 0$. Here,

$$\begin{aligned} \det(M - \lambda I) &= \begin{vmatrix} 6-\lambda & 0 \\ 0 & 7-\lambda \end{vmatrix} \\ &= (6-\lambda)(7-\lambda). \end{aligned} \tag{2.2}$$

Therefore,

$$(6-\lambda)(7-\lambda) = 0 \iff \lambda = 6 \text{ or } \lambda = 7. \tag{2.3}$$

Hence, $\text{eigs}(M) = \{6, 7\}$. Since M is real symmetric and both eigenvalues are strictly positive, it is *positive definite (PD)* by Definition 1.1.18.

Remark 2.1.3. The eigenvalues of a diagonal matrix are precisely its diagonal entries.

Now consider the following matrix:

$$\begin{bmatrix} 1 & -2 \\ 1 & -1 \end{bmatrix}. \tag{2.4}$$

To find its eigenvalues, we solve

$$\begin{aligned} \det(A - \lambda I) &= \begin{vmatrix} 1-\lambda & -2 \\ 1 & -1-\lambda \end{vmatrix} \\ &= (1-\lambda)(-1-\lambda) + 2 \\ &= -1 - \lambda + \lambda + \lambda^2 + 2 \\ &= \lambda^2 + 1 = 0. \end{aligned}$$

Therefore,

$$\lambda^2 = -1 \implies \lambda = \pm i.$$

Its eigenvalues $\pm i$ are not real. Therefore, this matrix does not fall into any class given in Definition 1.1.18. But there are matrices that guaranteed to have real eigenvalues, which we discuss next. Recall:

Definition 2.1.4 (Transpose of a matrix)**.** Let A be an $m \times n$ matrix. Denote its entry whose row number is i and column number is j by $a_{i,j}$. Then transpose of the matrix A, denoted by A^T, is an $n \times m$ matrix whose entry at j-th row and i-th column is $a_{i,j}$. Symbolically, if

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} & \dots & a_{1,n} \\ a_{2,1} & a_{2,2} & \dots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & \dots & a_{m,n} \end{bmatrix}, \tag{2.5}$$

then

$$A^T = \begin{bmatrix} a_{1,1} & a_{2,1} & \dots & a_{m,1} \\ a_{1,2} & a_{2,2} & \dots & a_{m,2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1,n} & a_{2,n} & \dots & a_{m,n} \end{bmatrix}. \tag{2.6}$$

Definition 2.1.5 (Symmetric matrix)**.** We call an $n \times n$ matrix A to be *symmetric* if

$$A = A^T. \tag{2.7}$$

Theorem 2.1.6

All the eigenvalues of a symmetric matrix A are real.

Proof. Will be done in the next class after developing some new idea. □

To prove this theorem, we now introduce new ideas in the context of vector spaces.

§2.1.1 Inner product and norm

It is generalisation of dot product of vectors that we learnt in the HSC/A-levels.

Example 2.1.7 (Motivating example)

Consider the $\mathbb{R}$-vector space $\mathbb{R}^2$. Define a function

$$\langle _, _ \rangle : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}, \tag{2.8}$$

given by

$$\langle _, _ \rangle (\mathbf{x}, \mathbf{y}) \coloneqq \mathbf{x}^T \mathbf{y}. \tag{2.9}$$

We will denote $\langle _, _ \rangle (\mathbf{x}, \mathbf{y})$ by $\langle \mathbf{x}, \mathbf{y} \rangle$.

Therefore, $\langle \mathbf{x}, \mathbf{y} \rangle$ is nothing but dot product of $\mathbf{x}$ and $\mathbf{y}$ in the high school sense, i.e.,

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x} \cdot \mathbf{y} \tag{2.10}$$

If $\mathbf{x} = \begin{bmatrix} x_1 & x_2 \end{bmatrix}^T$ and $\mathbf{y} = \begin{bmatrix} y_1 & y_2 \end{bmatrix}^T$, then

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x} \cdot \mathbf{y} = \sum_{k=1}^{2} x_k y_k. \tag{2.11}$$

We can then write length of a vector $\mathbf{x} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \in \mathbb{R}^2$, denoted by $\|\mathbf{x}\|$, in terms of dot product defined as above:

$$\|x\| = \sqrt{x_1^2 + x_2^2} = \sqrt{\langle x_1, x_2 \rangle}. \tag{2.12}$$

Furthermore, distance of $\mathbf{x}, \mathbf{y} \in \mathbb{R}^2$ can also be written in terms of the function $\langle _, _ \rangle$:

$$\begin{aligned}
&\sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2} && (2.13)\\
=&\sqrt{\langle (x_1 - y_1, x_2 - y_2), (x_1 - y_1, x_2 - y_2) \rangle} && (2.14)\\
=&\sqrt{\langle \mathbf{x} - \mathbf{y}, \mathbf{x} - \mathbf{y} \rangle} && (2.15)\\
=&\|\mathbf{x} - \mathbf{y}\|. && (2.16)
\end{aligned}$$

Notice, we previously did not have the notion of length of a vector or distance between two vectors in $\mathbb{R}^n$, for arbitrary n. But the above definition of dot product can be generalised to any $\mathbb{R}^n$, and using that we can meaningfully introduce the notion of length and distance in arbitrary $\mathbb{R}^n$. This generalised notion of dot product and length will be called *inner product* and *norm*, respectively.

Definition 2.1.8 (Inner product)**.** An *inner product* on a vector space V over $\mathbb{R}$ is a function

$$\langle \cdot, \cdot \rangle : V \times V \longrightarrow \mathbb{R} \tag{2.17}$$

which assigns a real number $\langle u, v \rangle$ to every pair of vectors $u, v \in V$, satisfying the following properties for all $u, v, w \in V$ and $c \in \mathbb{R}$:

1. **Linearity in the first argument:**

$$\langle cu + v, w \rangle = c\langle u, w \rangle + \langle v, w \rangle. \tag{2.18}$$

2. **Symmetry:**

$$\langle u, v \rangle = \langle v, u \rangle. \tag{2.19}$$

3. **Positive definiteness:**

$$\langle u, u \rangle \geq 0, \tag{2.20}$$

with equality if and only if $u = 0$.

A vector space equipped with an inner product is called an *inner product space*.

Example 2.1.9

In the vector space $\mathbb{R}^n$ define

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \mathbf{y}. \tag{2.21}$$

Then it can be checked that the definition above defines an inner product on $\mathbb{R}^n$.

Let $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$ and $c \in \mathbb{R}$. We verify the properties of an inner product.

1. **Linearity in the first argument:**

$$\begin{aligned}\langle c\mathbf{x} + \mathbf{y}, \mathbf{z}\rangle &= (c\mathbf{x} + \mathbf{y})^T\mathbf{z} \\ &= c\mathbf{x}^T\mathbf{z} + \mathbf{y}^T\mathbf{z} \\ &= c\langle \mathbf{x}, \mathbf{z}\rangle + \langle \mathbf{y}, \mathbf{z}\rangle.\end{aligned}$$

2. **Symmetry:**

$$\begin{aligned}\langle \mathbf{x}, \mathbf{y}\rangle &= \mathbf{x}^T\mathbf{y} \\ &= \sum_{i=1}^{n} x_i y_i \\ &= \sum_{i=1}^{n} y_i x_i \\ &= \mathbf{y}^T\mathbf{x} \\ &= \langle \mathbf{y}, \mathbf{x}\rangle.\end{aligned}$$

3. **Positive definiteness:**

$$\begin{aligned}\langle \mathbf{x}, \mathbf{x}\rangle &= \mathbf{x}^T\mathbf{x} \\ &= \sum_{i=1}^{n} x_i^2 \geq 0.\end{aligned}$$

Since $x_i^2 \geq 0$ for every i, equality holds if and only if $x_i = 0$ for all i, i.e., $\mathbf{x} = \mathbf{0}$.

Hence, the Euclidean dot product defines an inner product on $\mathbb{R}^n$.

Definition 2.1.10 (Norm induced by an inner product)**.** Suppose V is an $\mathbb{R}$-vector space with an inner product $\langle _, _\rangle$ defined on it. Then the function

$$\|.\| : V \to \mathbb{R} \tag{2.22}$$

defined by

$$\|\mathbf{x}\| \coloneqq \sqrt{\langle \mathbf{x}, \mathbf{x}\rangle} \tag{2.23}$$

is called *norm induced by this inner product.*

Example 2.1.11 (Euclidean norm on $\mathbb{R}^n$)
For $\mathbf{x} = (x_1, x_2, \ldots, x_n)^T \in \mathbb{R}^n$, the norm induced by the Euclidean inner product is

$$\begin{aligned}\|\mathbf{x}\|_2 &= \sqrt{\langle \mathbf{x}, \mathbf{x}\rangle} \\ &= \sqrt{\mathbf{x}^T\mathbf{x}}\end{aligned}$$

$$= \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}.$$

This is called the *Euclidean norm* or *2-norm* on $\mathbb{R}^n$.

Proposition 2.1.12

Norm induced by an inner product has some intrinsic properties:
For all $\mathbf{x}, \mathbf{y} \in V$ and $c \in \mathbb{R}$

1. **Positive definiteness:**
$$\|\mathbf{x}\| \geq 0, \tag{2.24}$$
and
$$\|\mathbf{x}\| = 0 \iff \mathbf{x} = 0. \tag{2.25}$$
2. **Homogeneity:**
$$\|c\mathbf{x}\| = |c| \, \|\mathbf{x}\|. \tag{2.26}$$
3. **Triangle inequality:**
$$\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|. \tag{2.27}$$

By making these properties definition, we can define general notion of norm that is independent from inner product.

Definition 2.1.13 (Norm on a vector space)**.** Let V be a real vector space. A *norm* on V is a function

$$\|\cdot\| : V \longrightarrow \mathbb{R}$$

that assigns a real number $\|v\|$ to each vector $v \in V$, satisfying the following properties for all $u, v \in V$ and all $c \in \mathbb{R}$:

1. **Non-negativity:**
$$\|v\| \geq 0. \tag{2.28}$$
2. **Definiteness:**
$$\|v\| = 0 \iff v = 0. \tag{2.29}$$
3. **Absolute homogeneity:**
$$\|cv\| = |c| \, \|v\|. \tag{2.30}$$
4. **Triangle inequality:**
$$\|u + v\| \leq \|u\| + \|v\|. \tag{2.31}$$

A real vector space V equipped with a norm is called a *normed vector space*.

Following example is particularly important for this course.

Example 2.1.14 (p-norm)

For $p \in \mathbb{R}$, define
$$\|.\|_p : \mathbb{R}^n \to \mathbb{R} \tag{2.32}$$

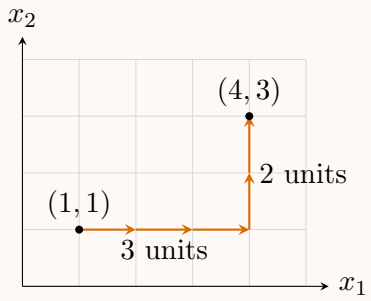
by

$$\|\mathbf{x}\|_p = (|x_1|^p + |x_2|^p + \cdots + |x_n|^p)^{\frac{1}{p}}, \tag{2.33}$$

where $\mathbf{x} = [x_1, \ldots, x_n]^T$.
We are not going to prove how this defines a norm on $\mathbb{R}^n$ though.
Putting $p = 2$ gives us the Euclidean norm of Example 2.1.11.
Putting $p = 1$ gives us

$$\|x\|_1 = |x_1| + |x_2| + \cdots + |x_n|, \tag{2.34}$$

which is also called "Manhattan norm" or "Taxicab" norm.
For example, consider the points[a] $\mathbf{x} = (1, 1)$ and $\mathbf{y} = (4, 3)$ in $\mathbb{R}^2$.



The Manhattan distance between these points is

$$d_1(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_1 = |4 - 1| + |3 - 1| = 5.$$

Letting $p \to \infty$ gives

$$\|\mathbf{x}\|_p = \max(|x_1|, |x_2|, \ldots, |x_n|) \tag{2.35}$$

because if $x_i = \max(|x_1|, |x_2|, \ldots, |x_n|)$, then

$$\begin{aligned}\lim_{p\to\infty} \|x\|_p &= \lim_{p\to\infty} (|x_i|^p)^{\frac{1}{p}} \left(\left|\frac{x_1}{x_i}\right|^p + \left|\frac{x_2}{x_i}\right|^p + \cdots + \left|\frac{x_i}{x_i}\right|^p + \cdots + \left|\frac{x_n}{x_i}\right|^p \right)^{\frac{1}{p}} \\ &= |x_i|(0 + 0 + \cdots + 1 + \cdots + 0)^0 \\ &= |x_i|.\end{aligned}$$

[a] vectors in $\mathbb{R}^n$ are also referred as "points in $\mathbb{R}^n$"

Question 2.1.15. Let $\mathbf{x} = \begin{bmatrix} -6 \\ 4 \\ 0 \\ 1 \end{bmatrix}$. Find the 1-norm and 3-norm of $\mathbf{x}$.

Answer 2.1.16. Using the definition of the p-norm, we have
For $p = 1$,

$$\begin{aligned}\|\mathbf{x}\|_1 &= |-6| + |4| + |0| + |1| \\ &= 6 + 4 + 0 + 1 \\ &= 11.\end{aligned}$$

For $p = 3$,

$$\begin{aligned}\|\mathbf{x}\|_3 &= \left(|-6|^3 + |4|^3 + |0|^3 + |1|^3\right)^{\frac{1}{3}} \\ &= (216 + 64 + 0 + 1)^{\frac{1}{3}} \\ &= \sqrt[3]{281}.\end{aligned}$$

§2.1.2 Metric

Definition 2.1.17 (Metric)**.** Let V be a nonempty set. A function

$$d : V \times V \to \mathbb{R} \tag{2.36}$$

is called a *metric* if for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$, the following properties hold:

1. **Positive definiteness:**
$$d(\mathbf{u}, \mathbf{v}) \geq 0, \tag{2.37}$$
with equality if and only if $\mathbf{u} = \mathbf{v}$.

2. **Symmetry:**
$$d(\mathbf{u}, \mathbf{v}) = d(\mathbf{v}, \mathbf{u}). \tag{2.38}$$

3. **Triangle inequality:**
$$d(\mathbf{u}, \mathbf{w}) \leq d(\mathbf{u}, \mathbf{v}) + d(\mathbf{v}, \mathbf{w}). \tag{2.39}$$

A set equipped with a metric is called a *metric space*.

Let V be a normed vector space. Then a metric can be induced by the norm as

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|. \tag{2.40}$$

In particular, if the norm is induced by an inner product, then

$$d(\mathbf{u}, \mathbf{v}) = \sqrt{\langle \mathbf{u} - \mathbf{v}, \mathbf{u} - \mathbf{v} \rangle}. \tag{2.41}$$

Example 2.1.18 (Euclidean distance on $\mathbb{R}^n$)
For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, the Euclidean distance is

$$d_2(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_2 = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}. \tag{2.42}$$

For example, if $\mathbf{x} = (1, 1)$ and $\mathbf{y} = (4, 3)$, then

$$\begin{aligned}d_2(\mathbf{x}, \mathbf{y}) &= \sqrt{(4-1)^2 + (3-1)^2} \\ &= \sqrt{13}.\end{aligned}$$

Definition 2.1.19 (Neighbourhood and ball)**.** Let d be a metric on $\mathbb{R}^n$. For $\mathbf{r}_0 \in \mathbb{R}^n$ and $a > 0$, the *open ball* of radius a centred at $\mathbf{r}_0$ is defined by

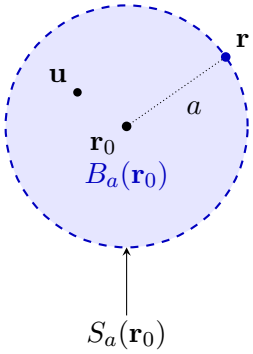
$$B_a(\mathbf{r}_0) = \{\mathbf{r} \in \mathbb{R}^n \mid d(\mathbf{r}, \mathbf{r}_0) < a\}. \tag{2.43}$$

It is also called the *a-neighbourhood* of $\mathbf{r}_0$.

The *sphere* of radius a centred at $\mathbf{r}_0$ is defined by

$$S_a(\mathbf{r}_0) = \{\mathbf{r} \in \mathbb{R}^n \mid d(\mathbf{r}, \mathbf{r}_0) = a\}. \tag{2.44}$$

In $\mathbb{R}^2$ with the Euclidean metric, the open ball consists of all points inside a circle, while the sphere is the circle itself. If $\mathbf{u} \in B_a(\mathbf{r}_0)$, then $d(\mathbf{u}, \mathbf{r}_0) < a$. If $\mathbf{r} \in S_a(\mathbf{r}_0)$, then $d(\mathbf{r}, \mathbf{r}_0) = a$.



3 Lecture 03

Date: 13/10/2026

§3.1 Review on linear algebra—continued

§3.1.1 Orthogonality

Definition 3.1.1 (Orthogonality)**.** Let V be an inner product space. Two vectors $\mathbf{x}, \mathbf{y} \in V$ are *orthogonal* if

$$\langle \mathbf{x}, \mathbf{y} \rangle = 0. \tag{3.1}$$

Example 3.1.2

Let $\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \in \mathbb{R}^3$. Then

$$\langle \mathbf{x}, \mathbf{y} \rangle = 1 \times 1 + 2 \times (-1) + 1 \times 1 = 0. \tag{3.2}$$

Therefore, $\mathbf{x}, \mathbf{y}$ are orthogonal/perpendicular.
In picture,

Add 3d picture of these orthogonal vectors.

But since inner product is defined for any $\mathbb{R}^n$, we can talk about orthogonality there as well.

Example 3.1.3

Let $\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \\ -7 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 1 \\ -1 \\ 1 \\ 5 \\ 0 \end{bmatrix} \in \mathbb{R}^3$. Then

$$\langle \mathbf{x}, \mathbf{y} \rangle = 1 \times 1 + 2 \times (-1) + 1 \times 1 + 0 \times 5 + (= 7) \times 0 = 0. \tag{3.3}$$

Therefore, $\mathbf{x}, \mathbf{y}$ are orthogonal.
Notice, $\mathbf{x}, \mathbf{y}$ are vectors in 5-dimensional space $\mathbb{R}^5$, which we can not visualise, yet we can talk about whether they are orthogonal/perpendicular to each other using the language of (linear) algebra.

If the basis vectors $e_1, e_2, \ldots, e_n$ of the vector space $\mathbb{R}^n$ are pairwise orthogonal, i.e.,

$$\langle e_i, e_j \rangle = 0, \qquad i \neq j, \tag{3.4}$$

then the coefficients of any vector $v \in \mathbb{R}^n$ with respect to this basis can be determined easily. Indeed, writing

$$v = \sum_{i=1}^{n} c_i e_i, \tag{3.5}$$

and taking the inner product with e_j on both sides, we obtain

$$\langle v, e_j \rangle = \left\langle \sum_{i=1}^{n} c_i e_i, e_j \right\rangle \tag{3.6}$$

$$= \sum_{i=1}^{n} c_i \langle e_i, e_j \rangle \tag{3.7}$$

$$= c_j \langle e_j, e_j \rangle, \tag{3.8}$$

since $\langle e_i, e_j \rangle = 0$ whenever $i \neq j$. Therefore,

$$c_j = \frac{\langle v, e_j \rangle}{\langle e_j, e_j \rangle} = \frac{\langle v, e_j \rangle}{\|e_j\|^2} \tag{3.9}$$

Thus, orthogonality allows us to determine each coefficient independently, without solving a system of linear equations. In particular, if the basis is orthonormal, then

$$c_j = \langle v, e_j \rangle \tag{3.10}$$

Let V be a normed space.

Definition 3.1.4 (Normalised vectors)**.** A vector $v \in V$ is called *normalised* if

$$\|v\| = 1. \tag{3.11}$$

Suppose $v \in V$ is not normalised. But

$$\frac{v}{\|v\|} \tag{3.12}$$

is normalised. Because

$$\left\| \frac{v}{\|v\|} \right\| = {}^{1} \frac{1}{\|v\|} \|v\| = 1. \tag{3.13}$$

The process making a vector normalised in this way, is called *normalisation*.

Definition 3.1.5 (Orthogonal basis)**.** A set of basis $\{e_1, e_2, \ldots, e_n\}$ of a vector space V is called *orthogonal basis* if

$$\langle e_i, e_j \rangle = \begin{cases} c \in \mathbb{R} & \text{if } i = j \\ 0 & \text{if } i \neq j. \end{cases} \tag{3.14}$$

Definition 3.1.6 (Orthonormal basis)**.** A set of basis $\{e_1, e_2, \ldots, e_n\}$ of a vector space V is called *orthonormal basis* if

$$\langle e_i, e_j \rangle = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j. \end{cases} \tag{3.15}$$

Notation (Kronecker/Dirac delta).

$$\delta_{ij} = \begin{cases} 1, i = j \\ 0, i \neq j. \end{cases} \tag{3.16}$$

Add examples related to Dirac delta

Now we prove Theorem 2.1.6.

Proof of Theorem 2.1.6. □

Complete the proof

In addition,

[1] Using absolute homogeneity property of norm

Theorem 3.1.7
Eigenvectors with different eigenvalues of a symmetric matrix are orthogonal.

Proof. □

Complete the proof.

Definition 3.1.8 (Anti/skew symmetric matrix)**.** A matrix A is called *anti/skew symmetric* if

$$A = -A^T. \tag{3.17}$$

Construction of symmetric matrices

Most matrices are neither symmetric nor anti symmetric. But given a matrix M, it can be written as sum of one symmetric and one anti symmetric matrix as follows:

Construction of anti-symmetric matrices

Complete

§3.2 Review on multivariable calculus

In this course, we will optimise functions whose inputs and outputs may consist of several real numbers. A function with several input variables is called a *multivariable function*. The branch of mathematics concerned with such functions is known as *multivariable calculus*.

§3.2.1 Terminology Related to Functions

Definition 3.2.1. Let A and B be nonempty sets. A function $f : A \to B$ is a rule that assigns to every element $x \in A$ exactly one element $f(x) \in B$. Here,

- A is called the *domain* of f.
- B is called the *codomain* of f.
- The *range*, or *image*, of f is the subset of B consisting of all values actually attained by f, namely $f(A) = \{f(x) : x \in A\}$.
- The *graph* of f is the set $\text{Graph}(f) = \{(x, f(x)) : x \in A\}$.

Example 3.2.2
Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^2$.
Its domain and codomain are both $\mathbb{R}$, but range is $[0, \infty)$.
The graph of f consists of all points of the form (x, x^2), where $x \in \mathbb{R}$. A portion of this graph is shown below.

add grpah of $f(x) = x^2$

Graph of a function $f : \mathbb{R} \to \mathbb{R}$ can be represented in a plane. But for functions of several variables, the situation is more complicated. For example, the graph of a function $f : \mathbb{R}^2 \to \mathbb{R}$ lies in $\mathbb{R}^3$ and can often be visualized as a surface. However, the graph of a function $f : \mathbb{R}^2 \to \mathbb{R}^2$ lies in $\mathbb{R}^4$, which cannot be represented directly in ordinary three-dimensional space.

Add graph of $x^2 + y$

§3.2.2 Directional Derivatives

The Derivative of a Single-Variable Function

We first recall the notion of differentiation from elementary calculus.

Definition 3.2.3. Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \to \mathbb{R}$, and let $a \in I$. We say that f is *differentiable at* a if the following limit exists as a finite real number:

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}. \tag{3.18}$$

The value $f'(a)$ is called the *derivative* of f at a.

Geometrically, the quotient $\frac{f(a+h)-f(a)}{h}$ represents the slope of the secant line joining $(a, f(a))$ and $(a+h, f(a+h))$. As h approaches zero, the secant slope approaches the slope of the tangent line, provided the limit exists.

Suppose now that $f : U \to \mathbb{R}$, where U is an open subset of $\mathbb{R}^n$ and $n \geq 2$.

Given $a \in U$, a small change in the input is represented by a vector $h \in \mathbb{R}^n$, rather than by a real number. One might attempt to imitate (3.18) by considering $\frac{f(a+h)-f(a)}{h}$.

However, this expression is not defined in the usual arithmetic of Euclidean spaces: its numerator is a real number, whereas its denominator is a vector. There is no canonical operation of division by a nonzero vector analogous to division by a nonzero real number.

There is also a geometric difficulty. In $\mathbb{R}$, a point can be approached only from the left or from the right. In $\mathbb{R}^n$, with $n \geq 2$, a point may be approached along infinitely many different directions. The rate of change of a function may depend on the particular direction in which its input is varied.

This leads us to the following definition:

Definition 3.2.4. Let $U \subseteq \mathbb{R}^n$ be open, let $f : U \to \mathbb{R}^m$, and let $a \in U$. For a nonzero vector $v \in \mathbb{R}^n$, the *directional derivative* of f at a in the direction v is defined by

$$D_v f(a) = \lim_{t \to 0} \frac{f(a+tv) - f(a)}{t}, \tag{3.19}$$

provided that this limit exists in $\mathbb{R}^m$.

The directional derivative $D_v f(a)$ measures the instantaneous rate of change of f at a as the input moves along the line $a + tv$. When v is a unit vector, it measures the rate of change per unit distance in that direction.

> **Remark 3.2.5.** The ordinary derivative is a special case of the directional derivative. Indeed, when $n = m = 1$ and $v = 1$, the definition reduces to $D_1 f(a) = f'(a)$.

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